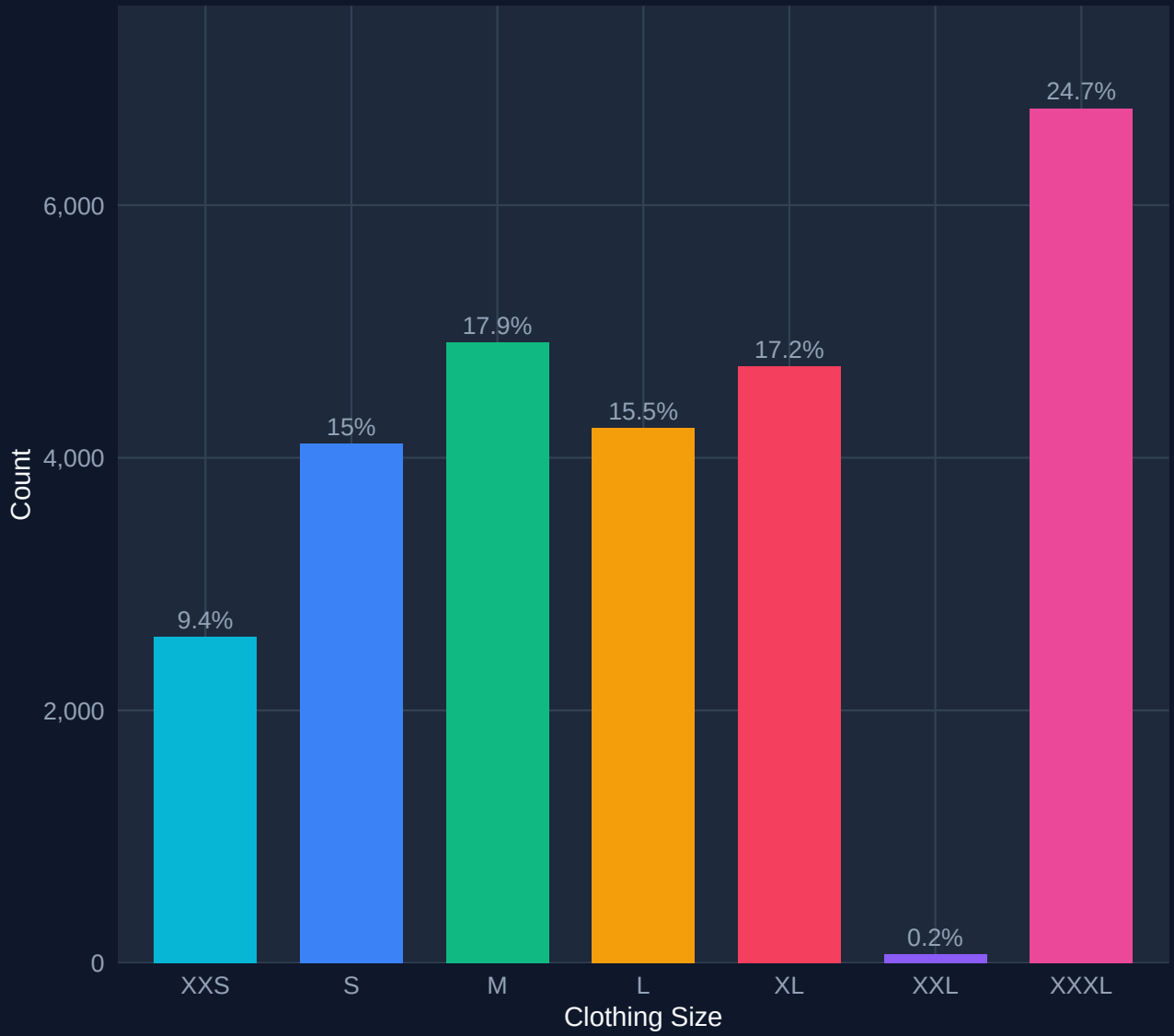
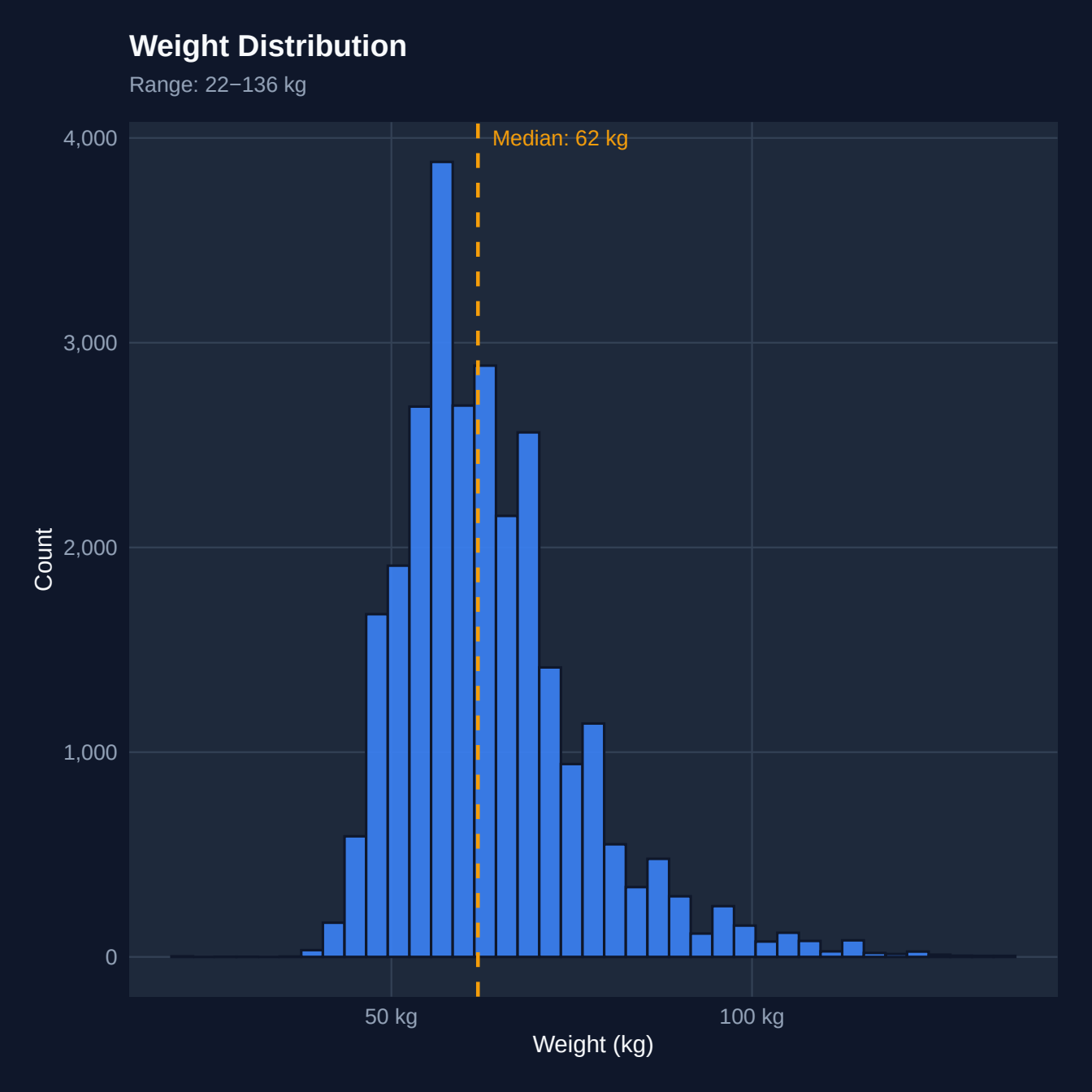
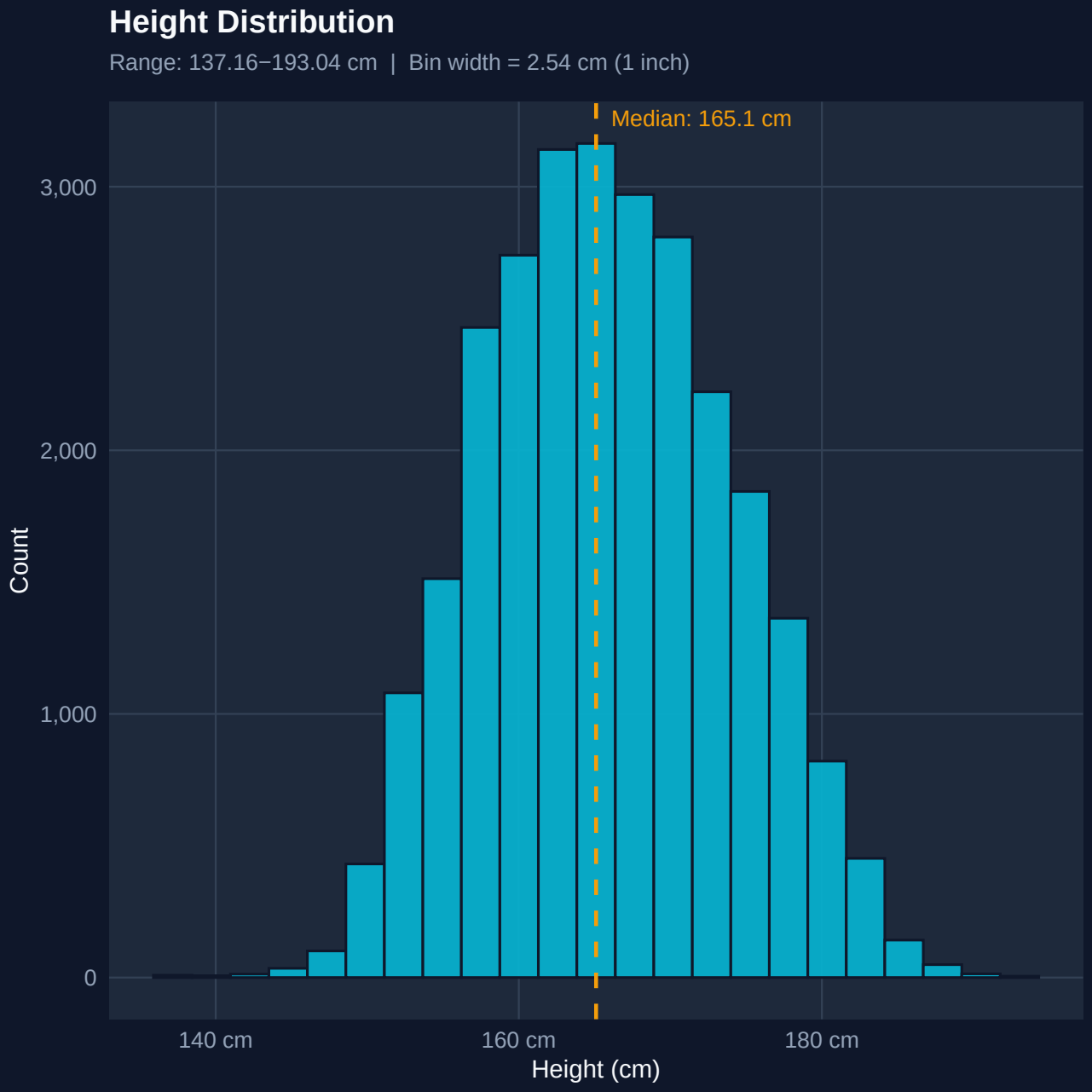
Size Class Distribution

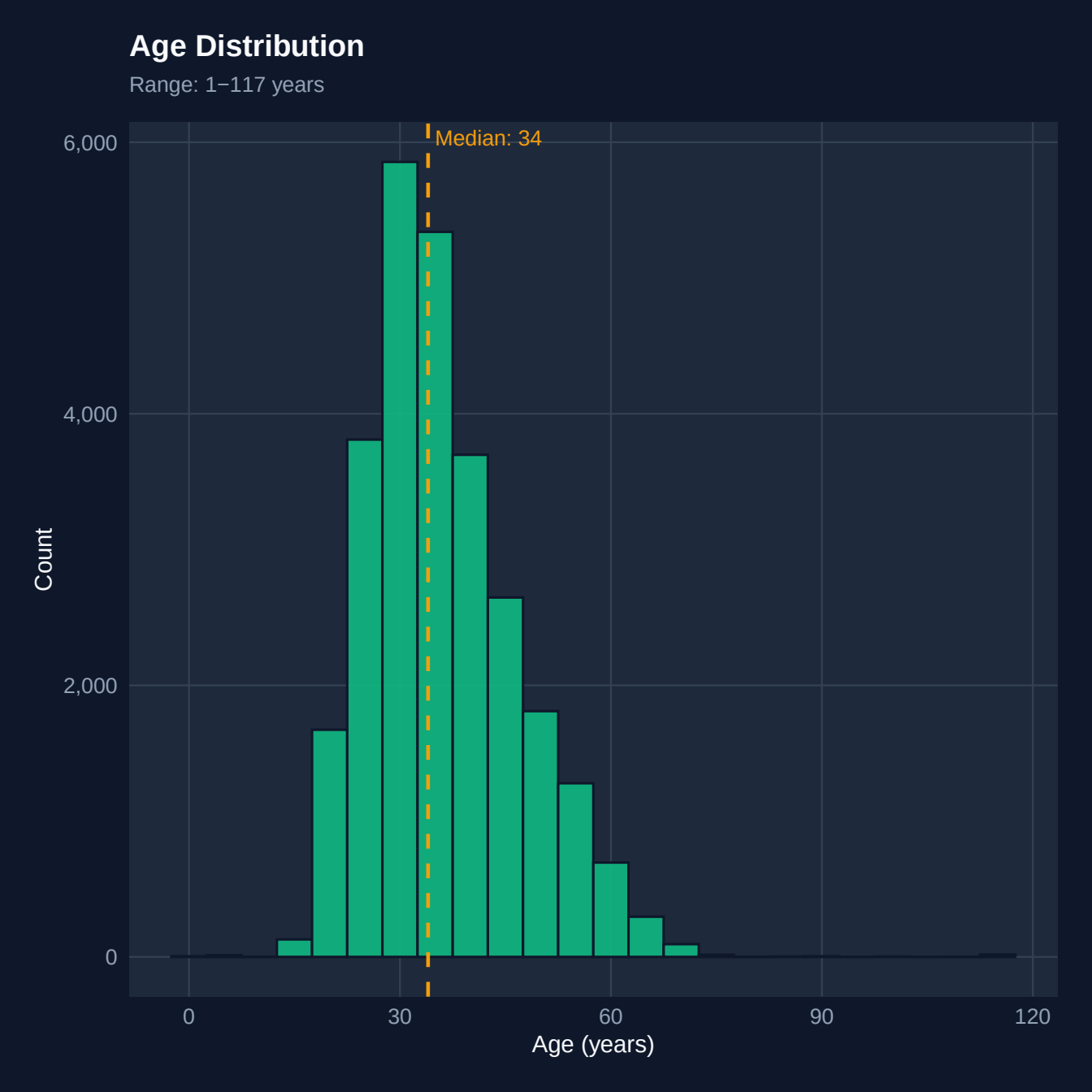
n = 27,391 records after cleaning



Note: XXL severely underrepresented (67 records)


Weight Distribution
Range: 22−136 kg
Median: 62 kg
4,000
3,000
2,000
1,000
0
Count
50 kg
100 kg
Weight (kg)


Height Distribution
Range: 137.16−193.04 cm | Bin width = 2.54 cm (1 inch)
Median: 165.1 cm
3,000
2,000
1,000
0
Count
140 cm
160 cm
180 cm
Height (cm)


Age Distribution
Range: 1−117 years
Median: 34
6,000
4,000
2,000
0
Count
0
30
60
90
120
Age (years)

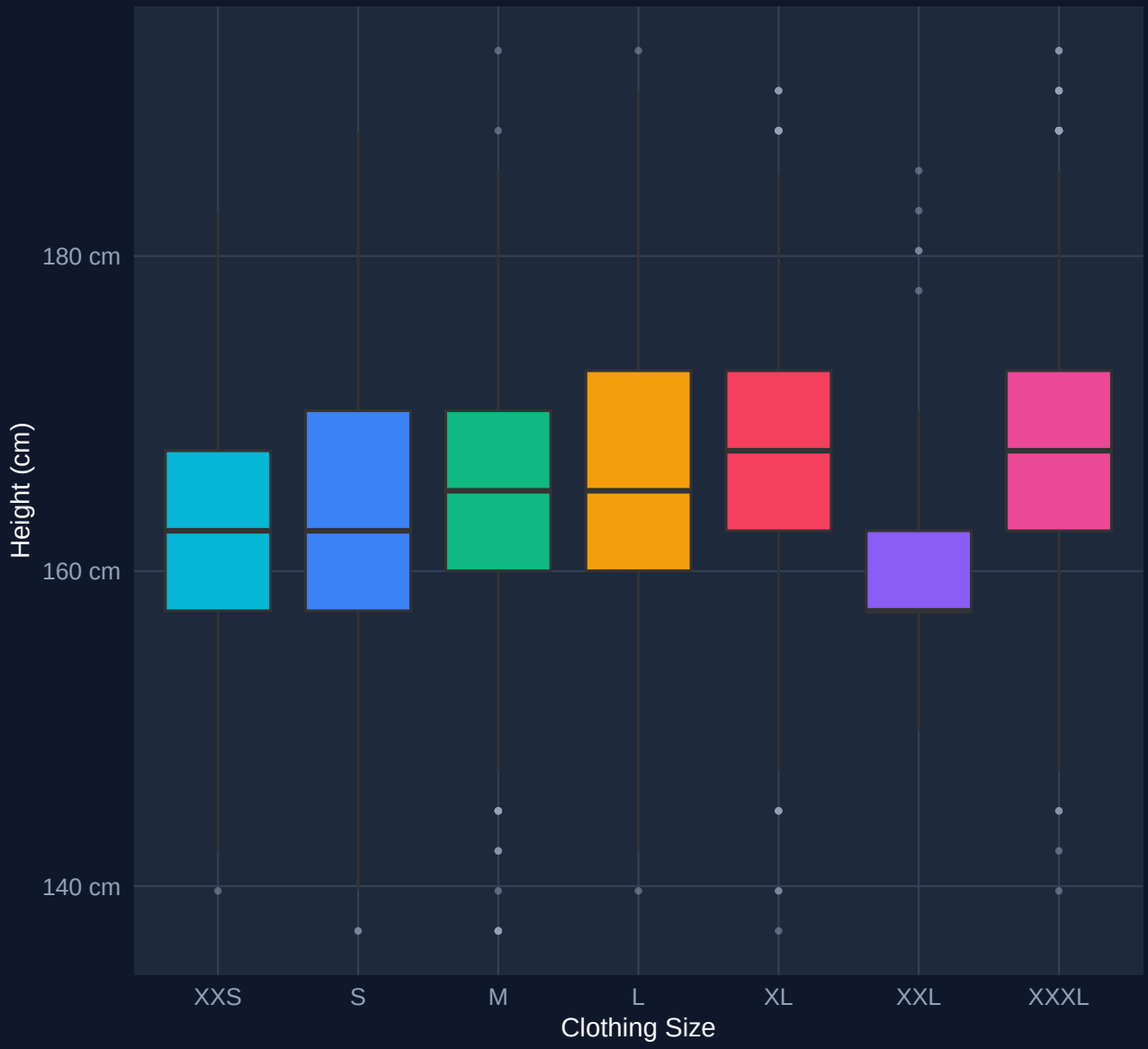
Weight by Clothing Size

Clear upward trend − heavier customers tend toward larger sizes



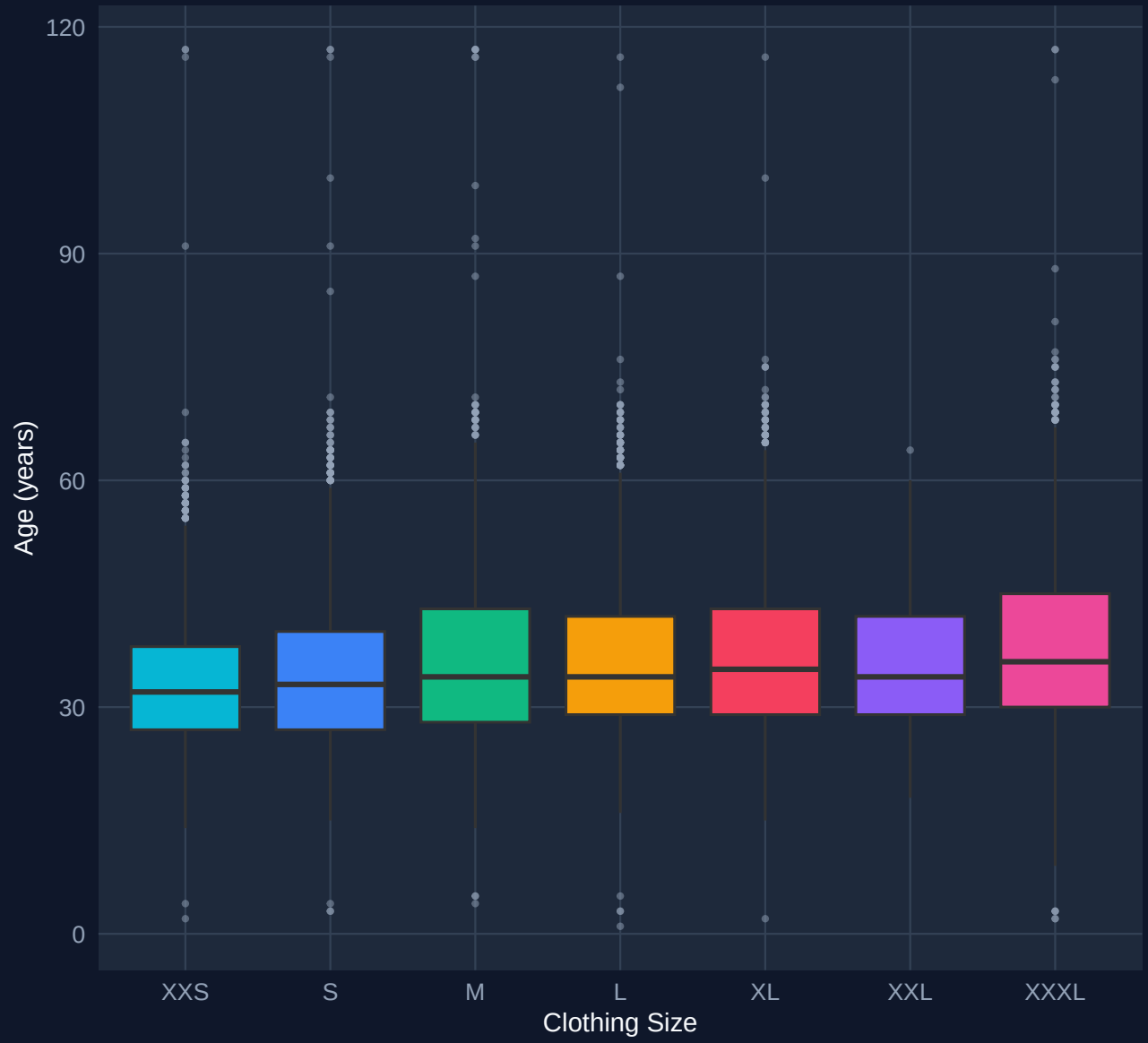
Height by Clothing Size

Taller customers tend toward larger sizes, though overlap is significant



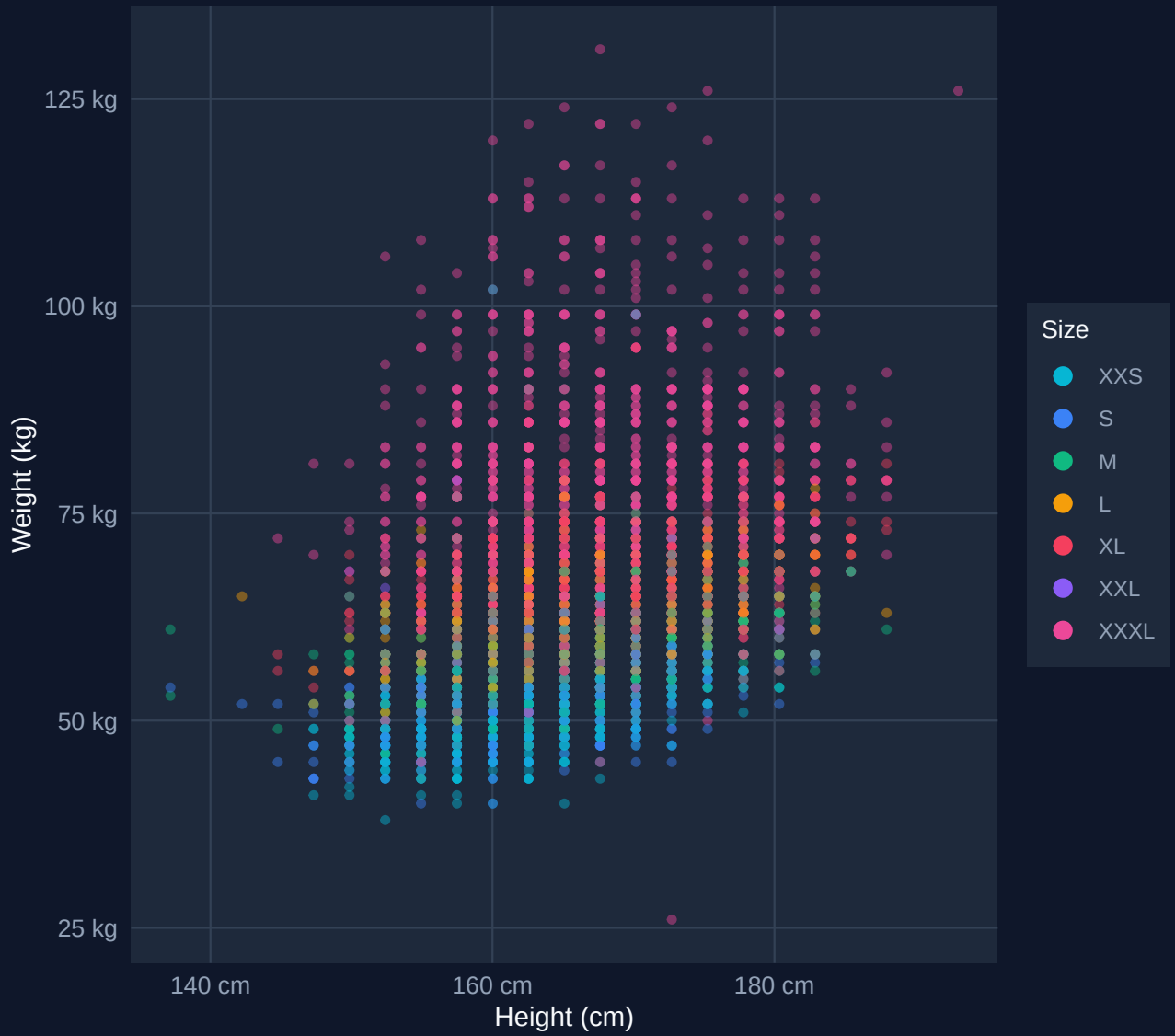
Age by Clothing Size

Age shows weak relationship with size − wide overlap across all classes



Weight vs Height by Size

Sample of 5,000 records − size bands visible but overlapping



Overlap confirms why a model is needed over a simple rule

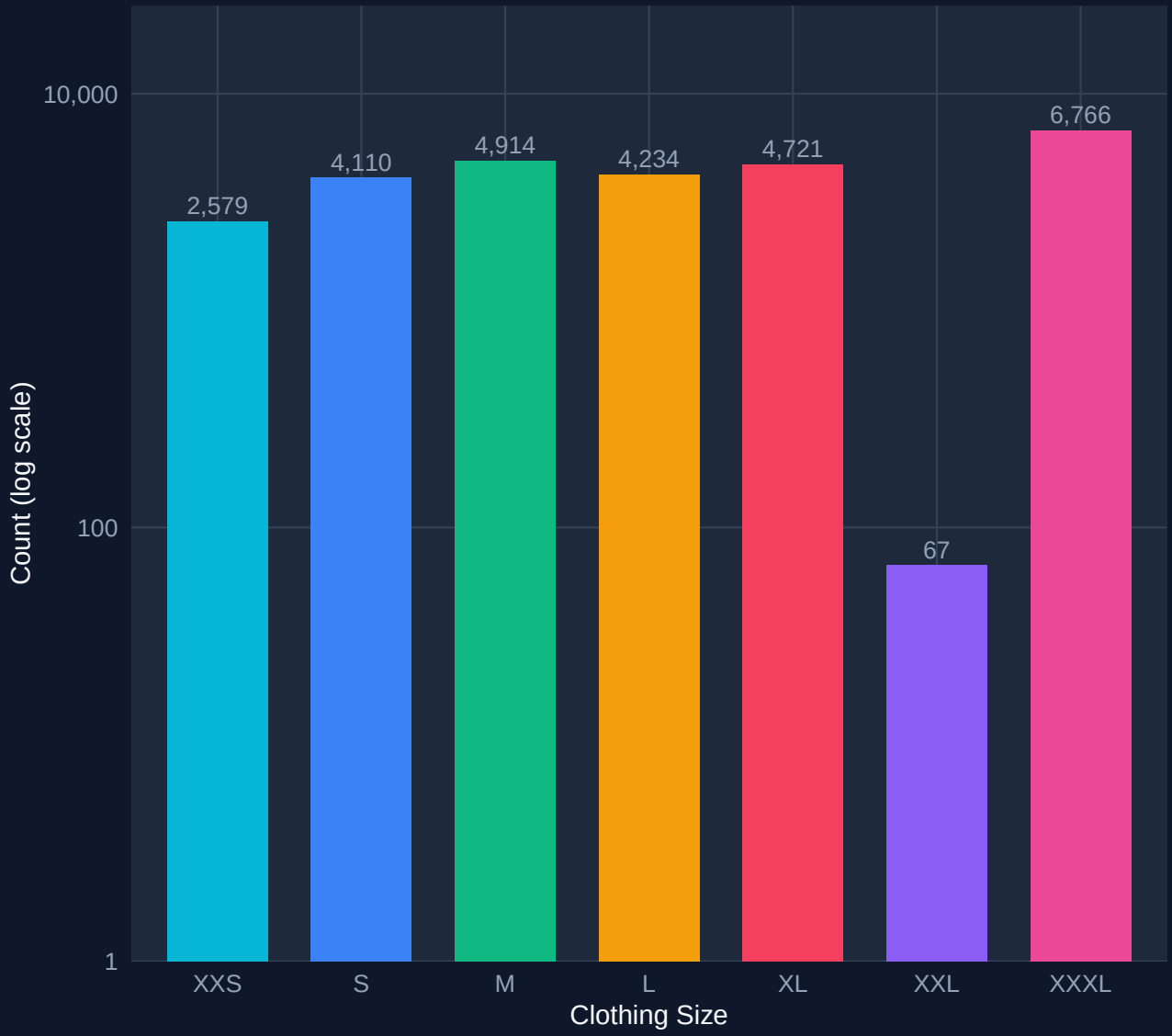
Predictor Correlation Matrix

Weight and height show moderate positive correlation



Class Imbalance − Log Scale

XXL has only 67 records vs XXXL with 6,766 − 101:1 ratio



Class weights will be applied in ranger to compensate